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(54) **LANGUAGE TRANSLATION WITH
HEAD-WORN COMPUTING**

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Apr. 24, 2017, now Pat. No. 10,146,772, which is a continuation of application No. 14/659,815, filed on Mar. 17, 2015, now Pat. No. 9,672,210, which is a continuation-in-part of application No. 14/490,586, filed on Sep. 18, 2014, now Pat. No. 9,423,842, which is a continuation-in-part of application No. 14/307,465, filed on Jun. 17, 2014, now Pat. No. 9,651,787, which is a continuation-in-part of application No. 14/323,123, filed on Jul. 3, 2014, now abandoned, which is a continuation-in-part of application No. 14/262,615, filed on Apr. 25, 2014, now Pat. No. 9,158,116.

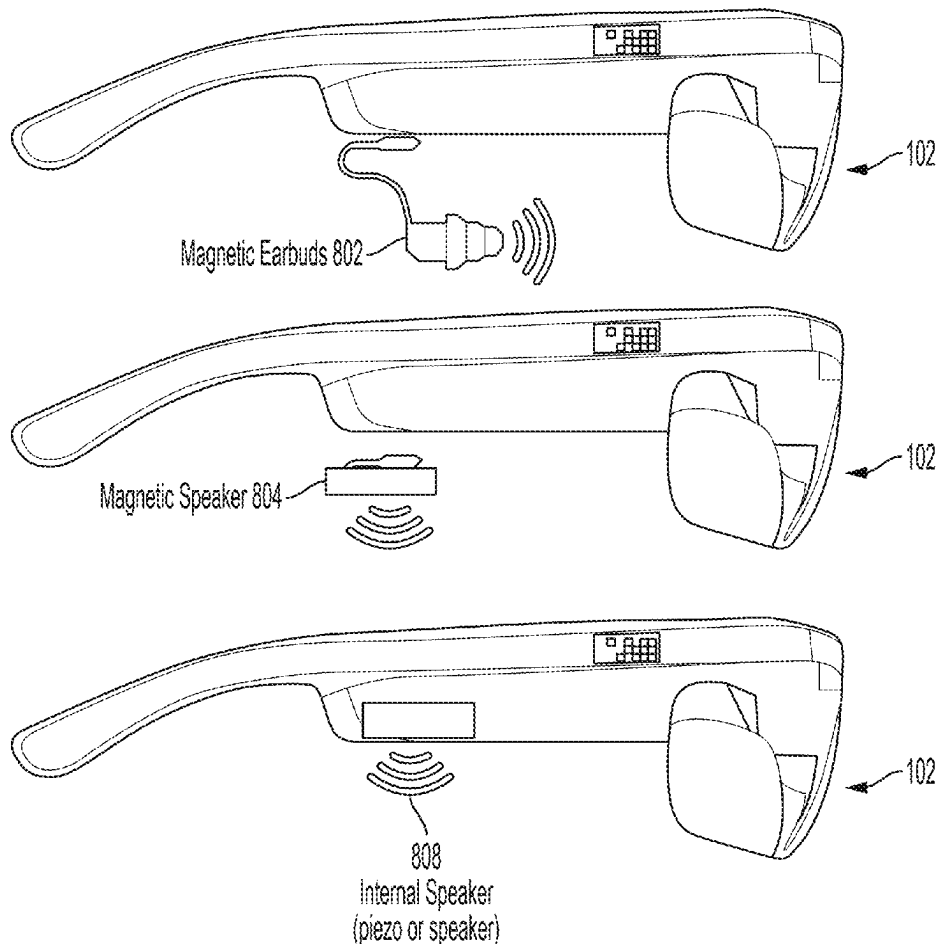
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2027/0156 (2013.01); **G02B 2027/0178**
(2013.01)

(57) **ABSTRACT**

Aspects of the present invention relate to language translation systems for head-worn computers.



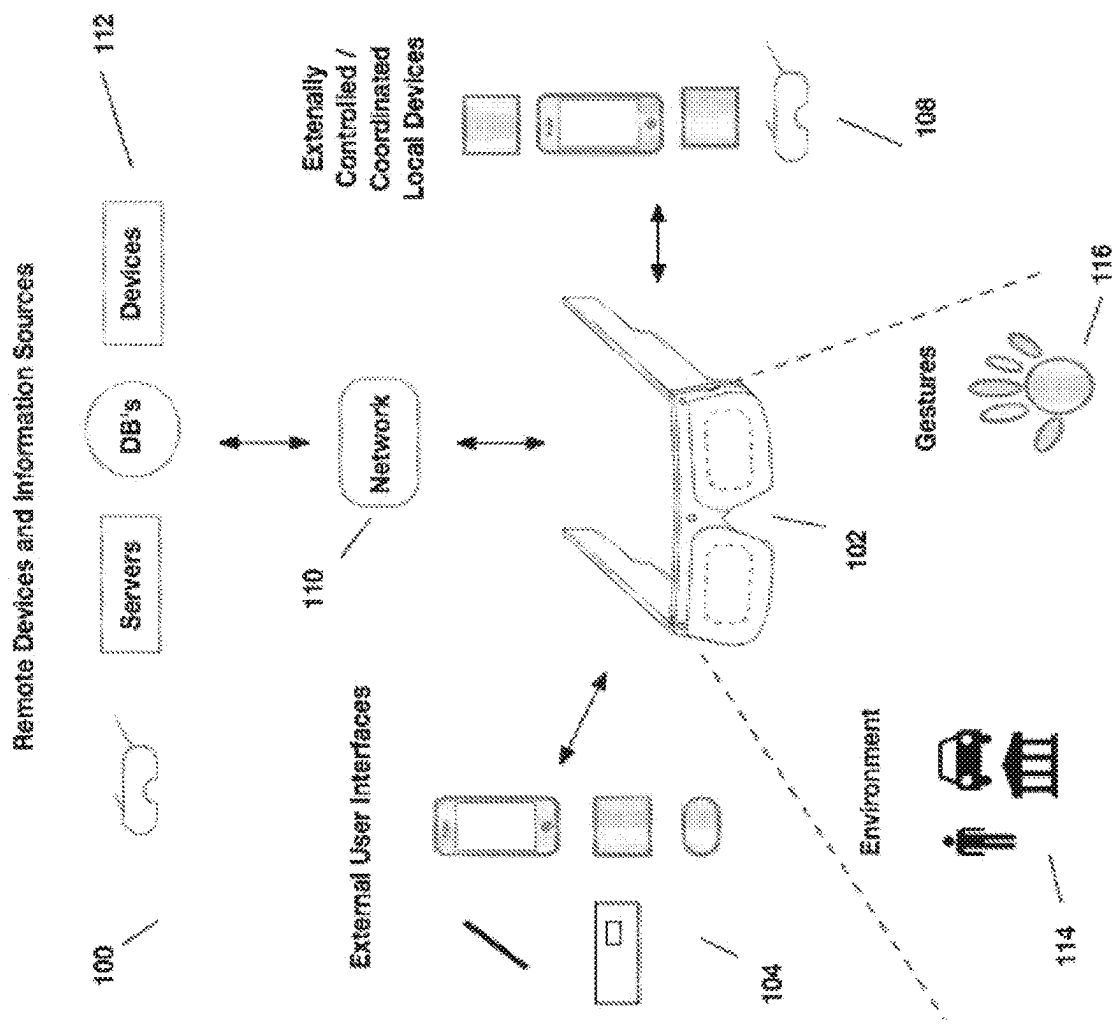


Figure 1

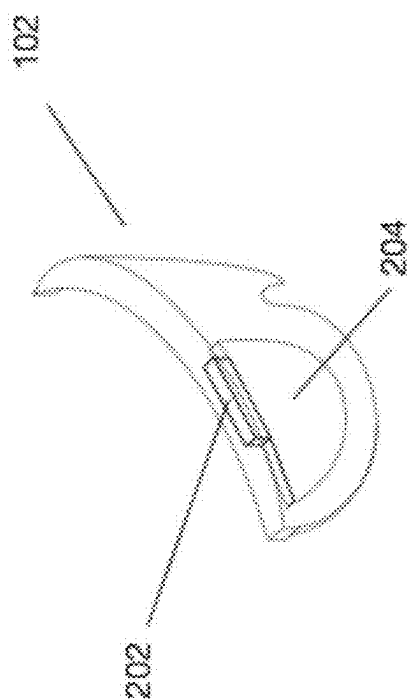
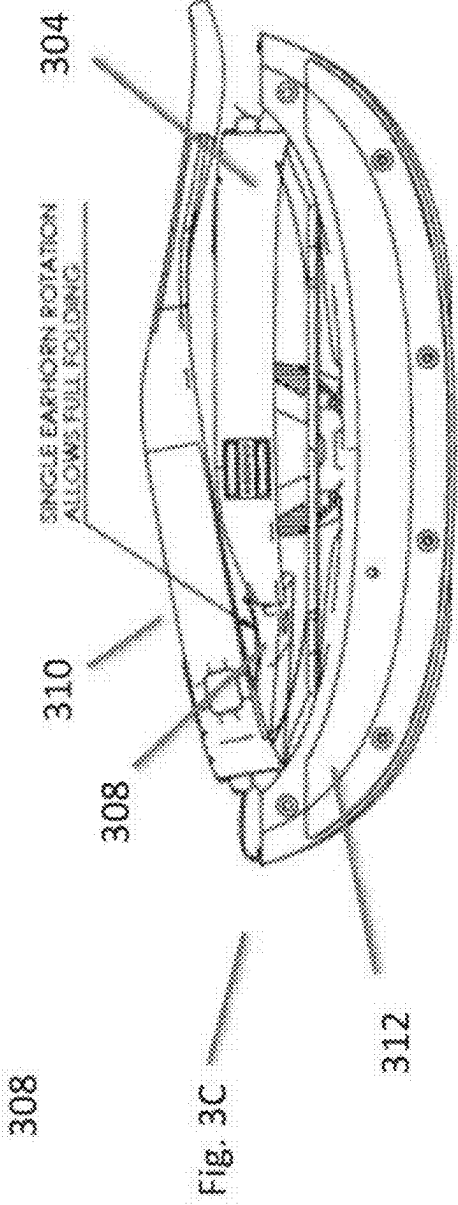
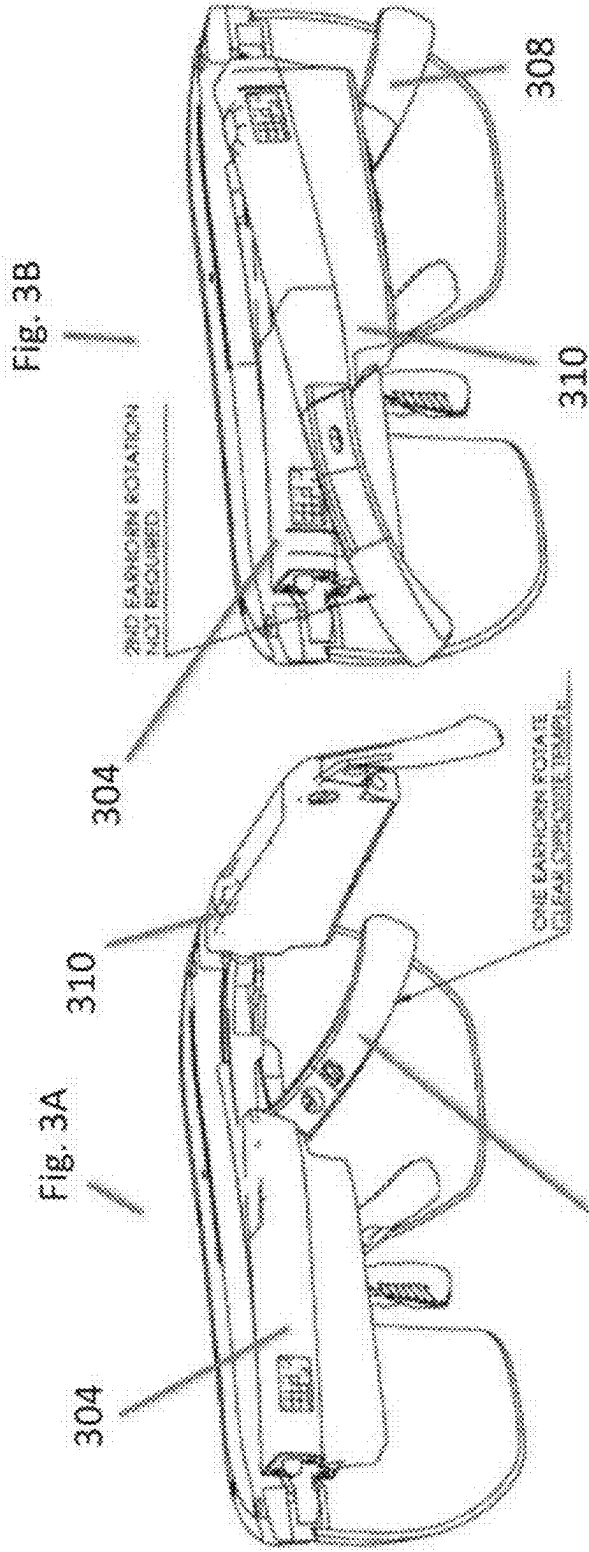


Figure 2



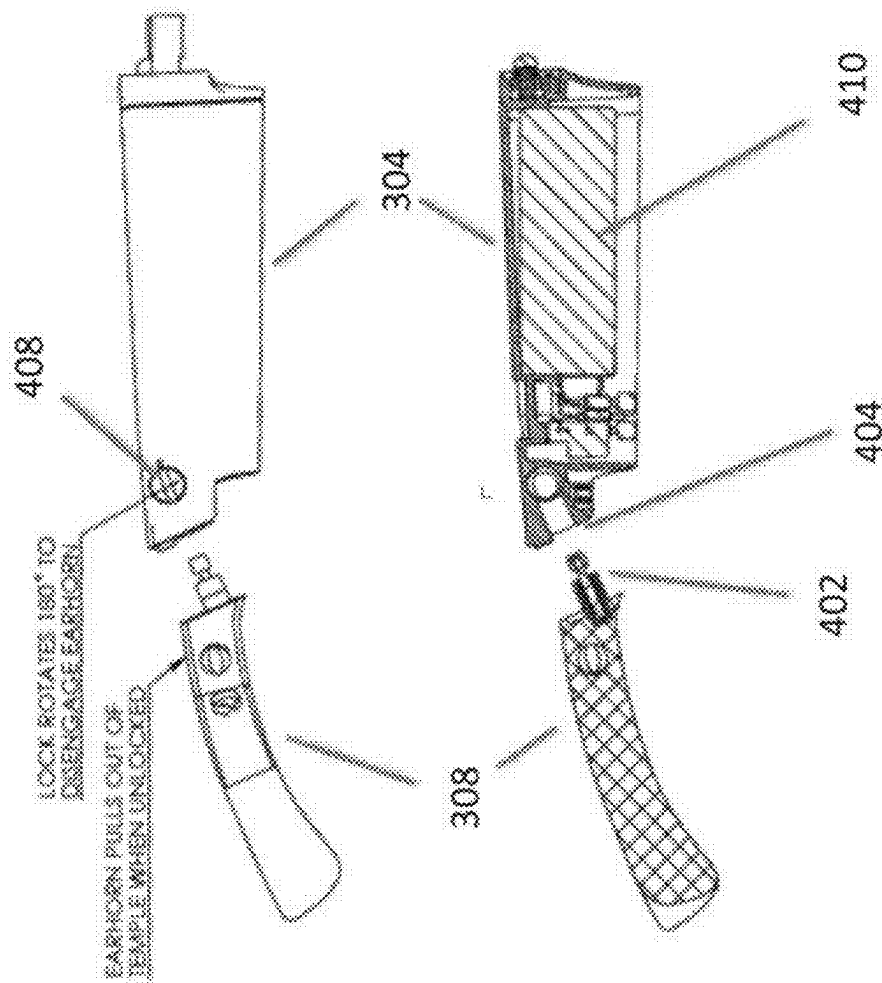


Figure 4

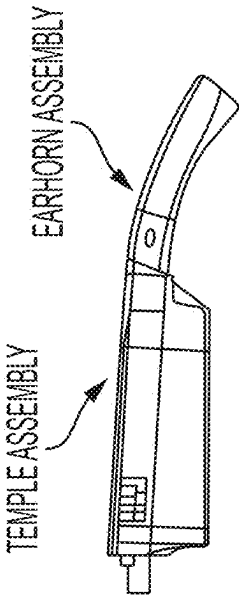


Fig. 5A

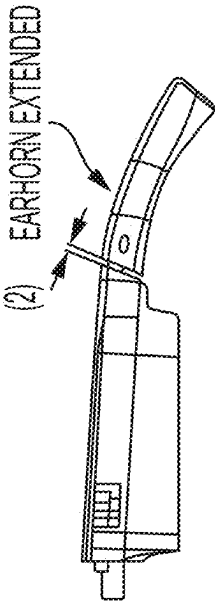


Fig. 5B

Fig. 5C

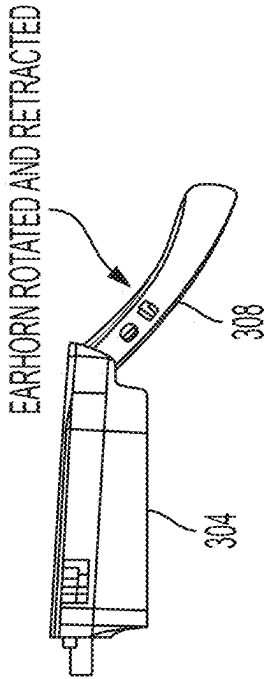


Fig. 5E

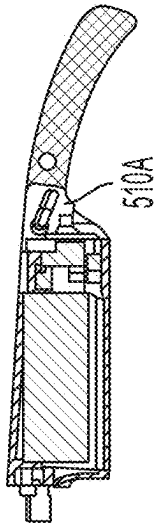


Fig. 5D

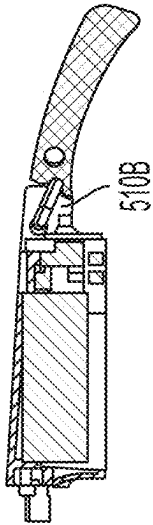
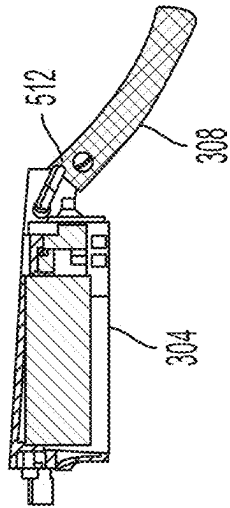


Fig. 5F



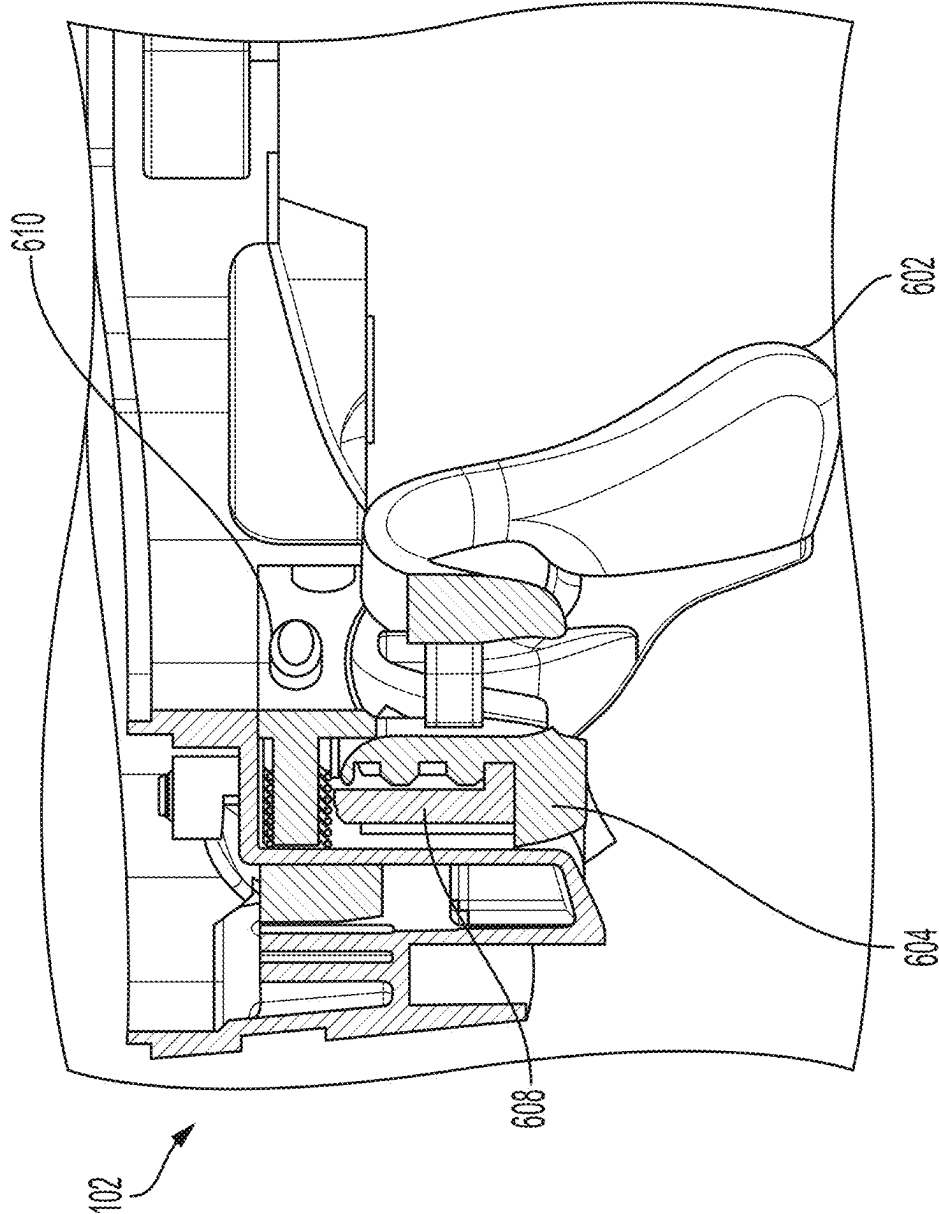
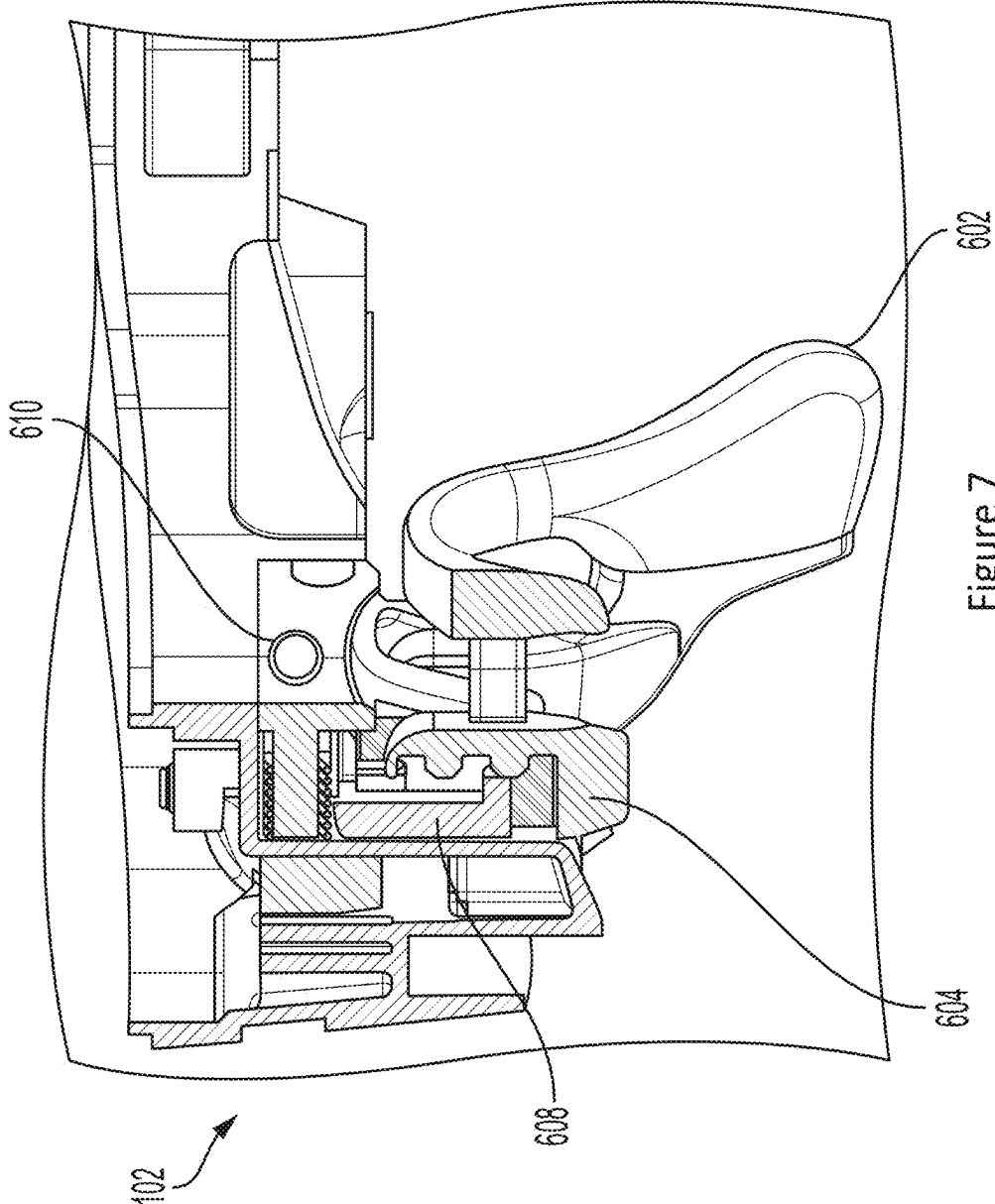


Figure 6



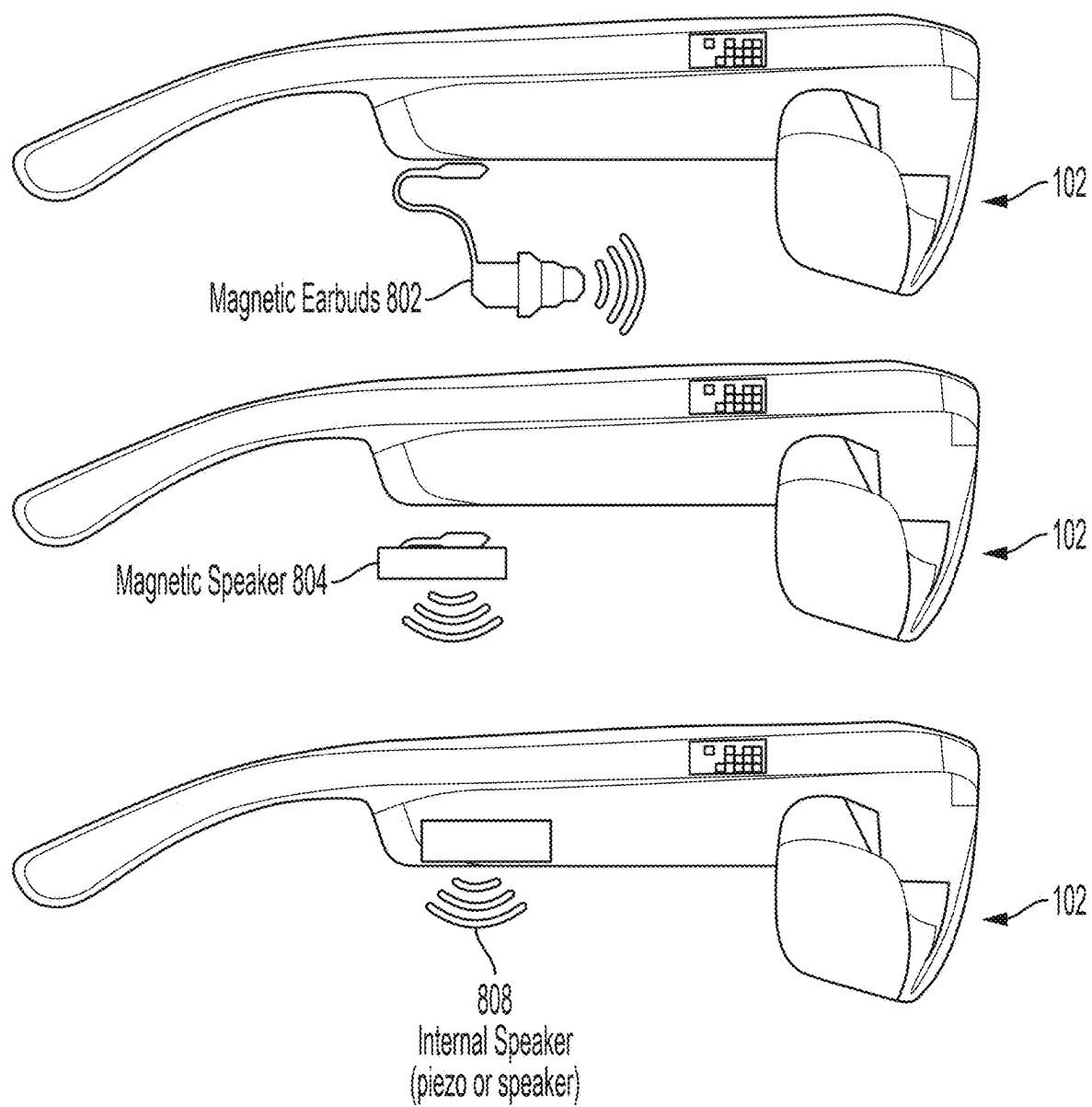


Figure 8

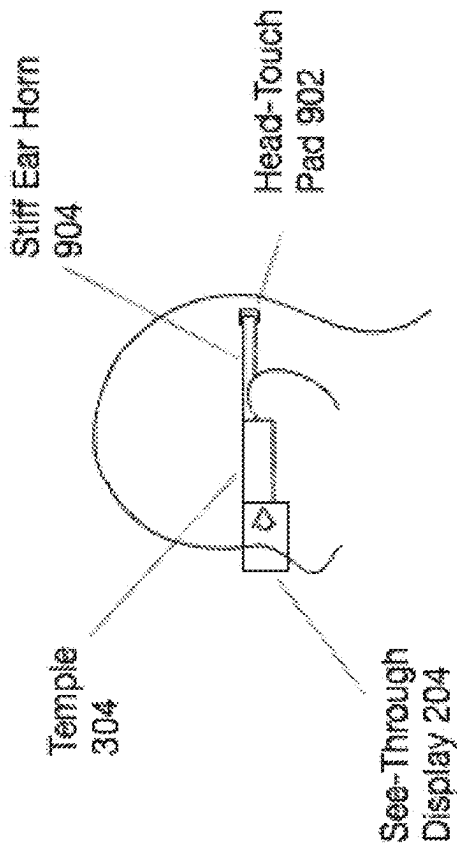


Figure 9

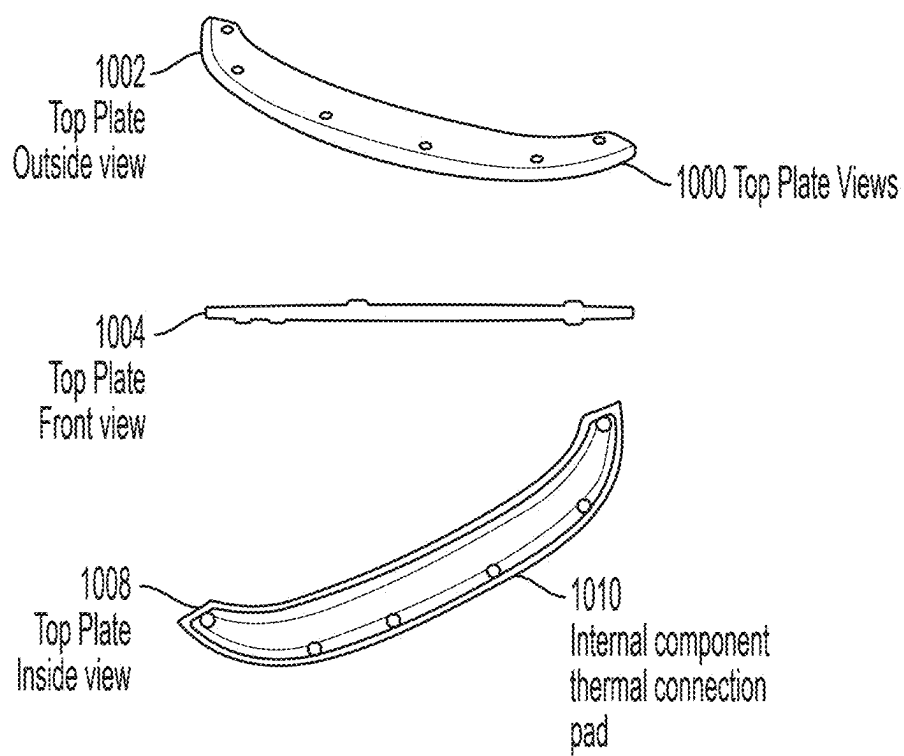


Figure 10

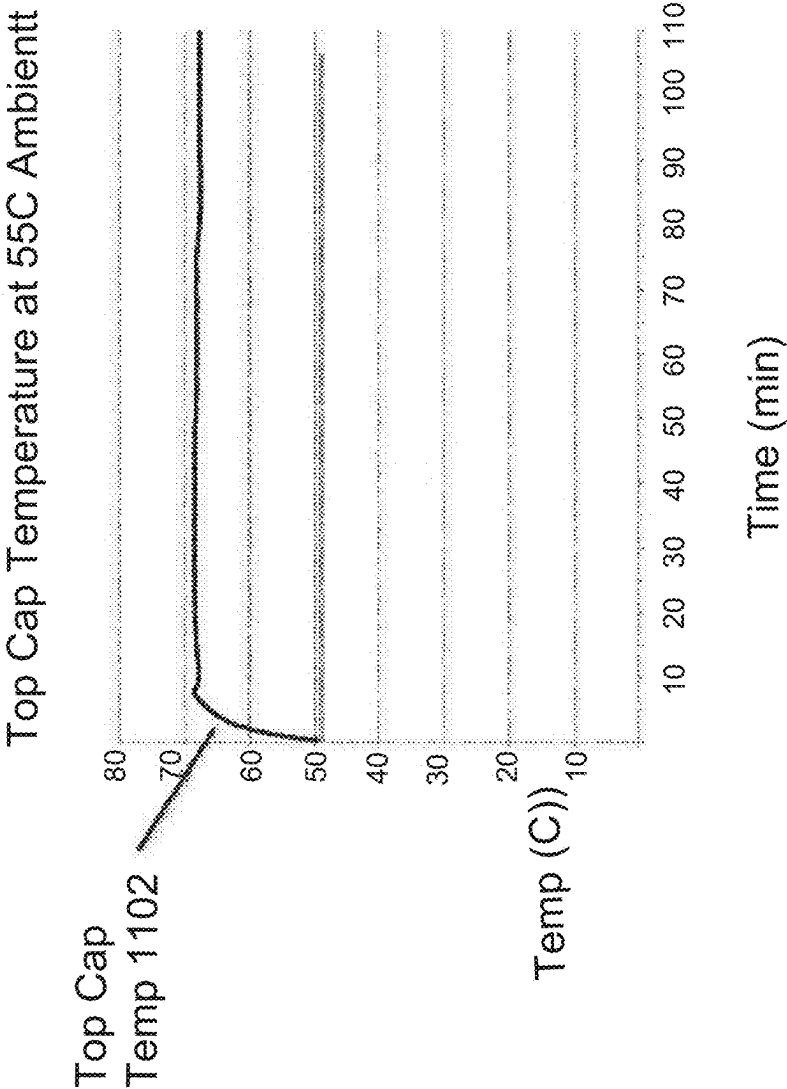


Figure 11

LANGUAGE TRANSLATION WITH HEAD-WORN COMPUTING

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/745,944, filed Jun. 17, 2024, which is a continuation of U.S. application Ser. No. 18/332,485, filed Jun. 9, 2023 (now U.S. Pat. No. 12,050,884, issued Jul. 30, 2024), which is continuation of U.S. application Ser. No. 17/083,228, filed Oct. 28, 2020 (now U.S. Pat. No. 11,727,223, issued Aug. 15, 2023), which is a continuation of U.S. application Ser. No. 16/149,405, filed Oct. 2, 2018 (now U.S. Pat. No. 10,853,589 issued Dec. 1, 2020), which is a continuation of U.S. application Ser. No. 15/494,827, filed Apr. 24, 2017 (now U.S. Pat. No. 10,146,772 issued Dec. 4, 2018), which is a continuation of U.S. application Ser. No. 14/659,815, filed Mar. 17, 2015 (now U.S. Pat. No. 9,672,210 issued Jun. 6, 2017), which is a continuation-in-part of the following U.S. Patent Applications:

[0002] U.S. application Ser. No. 14/490,586 filed Sep. 18, 2014 (now U.S. Pat. No. 9,423,842 issued Aug. 23, 2016); U.S. application Ser. No. 14/307,465 filed Jun. 17, 2014 (now U.S. Pat. No. 9,651,787 issued May 16, 2017); U.S. application Ser. No. 14/323,123 filed Jul. 3, 2014 (now abandoned); 14/262,615 filed Apr. 25, 2014 (now U.S. Pat. No. 9,158,116 issued Oct. 13, 2015).

[0003] All of the above-identified applications and patents are hereby incorporated by reference in their entirety.

BACKGROUND

Field of the Invention

[0004] This invention relates to head worn computing. More particularly, this invention relates to language translation systems for head-worn computers.

Description of Related Art

[0005] Wearable computing systems have been developed and are beginning to be commercialized. Many problems persist in the wearable computing field that need to be resolved to make them meet the demands of the market.

Summary

[0006] Aspects of the present invention relate to language translation systems for head-worn computers.

[0007] These and other systems, methods, objects, features, and advantages of the present invention will be apparent to those skilled in the art from the following detailed description of the preferred embodiment and the drawings. All documents mentioned herein are hereby incorporated in their entirety by reference.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Embodiments are described with reference to the following Figures. The same numbers may be used throughout to reference like features and components that are shown in the Figures:

[0009] FIG. 1 illustrates a head worn computing system in accordance with the principles of the present invention.

[0010] FIG. 2 illustrates a head worn computing system with optical system in accordance with the principles of the present invention.

[0011] FIG. 3A, FIG. 3B, and FIG. 3C illustrate three views of a head worn computer in accordance with the principles of the present invention.

[0012] FIG. 4 illustrates a temple and ear horn in accordance with the principles of the present invention.

[0013] FIG. 5A, FIG. 5B, FIG. 5C, FIG. 5D, FIG. 5E and FIG. 5F illustrate a temple and ear horn assembly in various states in accordance with the principles of the present invention.

[0014] FIG. 6 illustrates an adjustable nose bridge assembly in accordance with the principles of the present invention.

[0015] FIG. 7 illustrates an adjustable nose bridge assembly in accordance with the principles of the present invention.

[0016] FIG. 8 illustrates speaker assemblies for head-worn computers in accordance with the principles of the present invention.

[0017] FIG. 9 illustrates a stiff ear horn with a touch pad for a head-worn computer in accordance with the principles of the present invention.

[0018] FIG. 10 illustrates a top plate in accordance with the principles of the present invention.

[0019] FIG. 11 illustrates temperature test results of an embodiment of the present invention.

[0020] While the invention has been described in connection with certain preferred embodiments, other embodiments would be understood by one of ordinary skill in the art and are encompassed herein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT(S)

[0021] Aspects of the present invention relate to head-worn computing (“HWC”) systems. HWC involves, in some instances, a system that mimics the appearance of head-worn glasses or sunglasses. The glasses may be a fully developed computing platform, such as including computer displays presented in each of the lenses of the glasses to the eyes of the user. In embodiments, the lenses and displays may be configured to allow a person wearing the glasses to see the environment through the lenses while also seeing, simultaneously, digital imagery, which forms an overlaid image that is perceived by the person as a digitally augmented image of the environment, or augmented reality (“AR”).

[0022] HWC involves more than just placing a computing system on a person’s head. The system may need to be designed as a lightweight, compact and fully functional computer display, such as wherein the computer display includes a high resolution digital display that provides a high level of emersion comprised of the displayed digital content and the see-through view of the environmental surroundings. User interfaces and control systems suited to the HWC device may be required that are unlike those used for a more conventional computer such as a laptop. For the HWC and associated systems to be most effective, the glasses may be equipped with sensors to determine environmental conditions, geographic location, relative positioning to other points of interest, objects identified by imaging and movement by the user or other users in a connected group, and the like. The HWC may then change the mode of operation to match the conditions, location, positioning, movements, and

the like, in a method generally referred to as a contextually aware HWC. The glasses also may need to be connected, wirelessly or otherwise, to other systems either locally or through a network. Controlling the glasses may be achieved through the use of an external device, automatically through contextually gathered information, through user gestures captured by the glasses sensors, and the like. Each technique may be further refined depending on the software application being used in the glasses. The glasses may further be used to control or coordinate with external devices that are associated with the glasses.

[0023] Referring to FIG. 1, an overview of the HWC system 100 is presented. As shown, the HWC system 100 comprises a HWC 102, which in this instance is configured as glasses to be worn on the head with sensors such that the HWC 102 is aware of the objects and conditions in the environment 114. In this instance, the HWC 102 also receives and interprets control inputs such as gestures and movements 116 of body parts of a user. The HWC 102 may communicate with external user interfaces 104. The external user interfaces 104 may provide a physical user interface to take control instructions from a user of the HWC 102 and the external user interfaces 104 and the HWC 102 may communicate bi-directionally to affect the user's command and provide feedback to the external device 108. The HWC 102 may also communicate bi-directionally with externally controlled or coordinated local devices 108. For example, an external user interface 104 may be used in connection with the HWC 102 to control an externally controlled or coordinated local device 108. The externally controlled or coordinated local device 108 may provide feedback to the HWC 102 and a customized GUI may be presented in the HWC 102 based on the type of device or specifically identified device 108. The HWC 102 may also interact with remote devices and information sources 112 through a network connection 110. Again, the external user interface 104 may be used in connection with the HWC 102 to control or otherwise interact with any of the remote devices 108 and information sources 112 in a similar way as when the external user interfaces 104 are used to control or otherwise interact with the externally controlled or coordinated local devices 108. Similarly, HWC 102 may interpret gestures 116 (e.g. captured from forward, downward, upward, rearward facing sensors such as camera(s), range finders, IR sensors, etc.) or environmental conditions sensed in the environment 114 to control either local or remote devices 108 or 112.

[0024] We will now describe each of the main elements depicted on FIG. 1 in more detail; however, these descriptions are intended to provide general guidance and should not be construed as limiting. Additional description of each element may also be further described herein.

[0025] The HWC 102 is a computing platform intended to be worn on a person's head. The HWC 102 may take many different forms to fit many different functional requirements. In some situations, the HWC 102 will be designed in the form of conventional glasses. The glasses may or may not have active computer graphics displays. In situations where the HWC 102 has integrated computer displays the displays may be configured as see-through displays such that the digital imagery can be overlaid with respect to the user's view of the environment 114. There are a number of see-through optical designs that may be used, including ones that have a reflective display (e.g. LCoS, DLP), emissive displays (e.g. OLED, LED), hologram, TIR waveguides, and

the like. In embodiments, lighting systems used in connection with the display optics may be solid state lighting systems, such as LED, OLED, quantum dot, quantum dot LED, etc. In addition, the optical configuration may be monocular or binocular. It may also include vision corrective optical components. In embodiments, the optics may be packaged as contact lenses. In other embodiments, the HWC 102 may be in the form of a helmet with a see-through shield, sunglasses, safety glasses, goggles, a mask, fire helmet with see-through shield, police helmet with see through shield, military helmet with see-through shield, utility form customized to a certain work task (e.g. inventory control, logistics, repair, maintenance, etc.), and the like.

[0026] The HWC 102 may also have a number of integrated computing facilities, such as an integrated processor, integrated power management, communication structures (e.g. cell net, WiFi, Bluetooth, local area connections, mesh connections, remote connections (e.g. client server, etc.)), and the like. The HWC 102 may also have a number of positional awareness sensors, such as GPS, electronic compass, altimeter, tilt sensor, IMU, and the like. It may also have other sensors such as a camera, rangefinder, hyperspectral camera, Geiger counter, microphone, spectral illumination detector, temperature sensor, chemical sensor, biologic sensor, moisture sensor, ultrasonic sensor, and the like.

[0027] The HWC 102 may also have integrated control technologies. The integrated control technologies may be contextual based control, passive control, active control, user control, and the like. For example, the HWC 102 may have an integrated sensor (e.g. camera) that captures user hand or body gestures 116 such that the integrated processing system can interpret the gestures and generate control commands for the HWC 102. In another example, the HWC 102 may have sensors that detect movement (e.g. a nod, head shake, and the like) including accelerometers, gyros and other inertial measurements, where the integrated processor may interpret the movement and generate a control command in response. The HWC 102 may also automatically control itself based on measured or perceived environmental conditions. For example, if it is bright in the environment the HWC 102 may increase the brightness or contrast of the displayed image. In embodiments, the integrated control technologies may be mounted on the HWC 102 such that a user can interact with it directly. For example, the HWC 102 may have a button(s), touch capacitive interface, and the like.

[0028] As described herein, the HWC 102 may be in communication with external user interfaces 104. The external user interfaces may come in many different forms. For example, a cell phone screen may be adapted to take user input for control of an aspect of the HWC 102. The external user interface may be a dedicated UI, such as a keyboard, touch surface, button(s), joy stick, and the like. In embodiments, the external controller may be integrated into another device such as a ring, watch, bike, car, and the like. In each case, the external user interface 104 may include sensors (e.g. IMU, accelerometers, compass, altimeter, and the like) to provide additional input for controlling the HWC 102.

[0029] As described herein, the HWC 102 may control or coordinate with other local devices 108. The external devices 108 may be an audio device, visual device, vehicle, cell phone, computer, and the like. For instance, the local

external device **108** may be another HWC **102**, where information may then be exchanged between the separate HWCs **108**.

[0030] Similar to the way the HWC **102** may control or coordinate with local devices **106**, the HWC **102** may control or coordinate with remote devices **112**, such as the HWC **102** communicating with the remote devices **112** through a network **110**. Again, the form of the remote device **112** may have many forms. Included in these forms is another HWC **102**. For example, each HWC **102** may communicate its GPS position such that all the HWCs **102** know where all of HWC **102** are located.

[0031] FIG. **2** illustrates a HWC **102** with an optical system that includes an upper optical module **202** and a lower optical module **204**. While the upper and lower optical modules **202** and **204** will generally be described as separate modules, it should be understood that this is illustrative only and the present invention includes other physical configurations, such as that when the two modules are combined into a single module or where the elements making up the two modules are configured into more than two modules. In embodiments, the upper module **202** includes a computer controlled display (e.g. LCoS, DLP, OLED, etc.) and image light delivery optics. In embodiments, the lower module includes eye delivery optics that are configured to receive the upper module's image light and deliver the image light to the eye of a wearer of the HWC. In FIG. **2**, it should be noted that while the upper and lower optical modules **202** and **204** are illustrated in one side of the HWC such that image light can be delivered to one eye of the wearer, that it is envisioned by the present invention that embodiments will contain two image light delivery systems, one for each eye. It should also be noted that while many embodiments refer to the optical modules as "upper" and "lower" it should be understood that this convention is being used to make it easier for the reader and that the modules are not necessarily located in an upper-lower relationship. For example, the image generation module may be located above the eye delivery optics, below the eye delivery optics, on a side of the eye delivery optics, or otherwise positioned to satisfy the needs of the situation and/or the HWC **102** mechanical and optical requirements.

[0032] An aspect of the present invention relates to the mechanical and electrical construction of a side arm of a head worn computer. In general, when a head worn computer takes the form of glasses, sun-glasses, certain goggles, or other such forms, two side arms are included for mounting and securing the head worn computer on the ears of a person wearing the head worn computer. In embodiments, the side arms may also contain electronics, batteries, wires, antennas, computer processors, computer boards, etc. In embodiments, the side arm may include two or more sub assemblies. For example, as will be discussed in more detail below, the side arm may include a temple section and an ear horn section. The two sections may, for example, be mechanically arranged to allow an ear horn section to move such that both side arms can fold into a closed position.

[0033] FIG. **3A**, FIG. **3B** and FIG. **3C** illustrate three separate views of a head worn computer **102** according to the principles of the present invention. Turning to the head worn computer illustrated as FIG. **3A**, one side arm of the HWC **102** is folded into its closed position. The ear horn section **308** of the side arm is rotated relative to its temple section **304** to create space relative to the other side arm **310**

so when the other side arm is moved into its closed position it can fully close. In a situation where the ear horn did not rotate to create the space (not illustrated) the ear horn would physically interfere with the other side arm **310**, when the side arm was in the closed position, and prevent the other side arm **310** from fully closing. The HWC FIG. **3B** view illustrates the HWC FIG. **3B** with both side arms folded into a fully closed position. Note that the ear horn **308** is in the rotated position with respect to its temple section **304** such that the other arm **310** closed without interfering with the ear horn **308**. The HWC FIG. **3C** view also illustrates both arms in closed positions with the ear horn **308** rotated to create the space for the other arm **310** to fully close. FIG. **3C** also illustrates a portion of the HWC **102** where electronics may be housed in a top mount **312**. The top mount may contain electronics, sensors, optics, processors, memory, radios, antennas, etc.

[0034] FIG. **4** illustrates a side arm configuration in accordance with the principles of the present invention. In this embodiment, the side arm includes two sub assemblies: the temple section **304** and the ear horn **308**. FIG. **4** illustrates two views of the side arm assembly, one from an outer perspective and one from a sectioned perspective. The ear horn includes a pin **402** that is designed to fit into a hole **404** and to be secured by connector **408**. The connector **408** is rotatable and in one position locks the pin **402** in place and in another position unsecures the pin **402** such that the ear horn **308** can be removed and re-attached to the temple section **304**. This allows the detachment and re-attachment of the ear horn **308** from the temple section **304**. This also allows for the sale of different ear horns **308** for replacement, of which a variety of colors and patterns may be offered. In embodiments, the temple section **304** may include a battery compartment **410** and other electronics, wires, sensors, processors, etc.

[0035] FIG. **5A**, FIG. **5B**, FIG. **5C**, FIG. **5D**, FIG. **5E** and FIG. **5F** illustrate several views of a HWC side arm with temple **304** and ear horn **308** sections. The views include outer perspectives and cross sections as well as various states of the security of the ear horn **308** with the temple section **304**. One embodiment of an outer perspective and cross-section of a temple assembly and earhorn assembly is shown in FIG. **5A** and FIG. **5B**, respectively, including connector and pin assembly **510A**, wherein the ear horn is in its final secured position and ready to be put on the head of a user FIG. **5C** and FIG. **5D** illustrate the ear horn **308** and the temple section **304** in a secure, but separated and un-rotated position. The same pin **402** and connector **408** system described in connection with FIG. **4** is illustrated in the cross sections of FIG. **5E** and FIG. **5F** at connector and pin assembly **512**. In the secured un-rotated position the pin is pulled internally within the temple section firmly such that it stays in place. FIG. **5C** and FIG. **5D** illustrate a state where the ear horn **308** is separated from the temple section **304**. This state is achieved when pressure is used to pull on the ear horn **308**. In embodiments, the pressure is exerted by a user pulling on the ear horn **308**, which compresses a spring in the connector and pin assembly **510B** that is mechanically associated with the pin **402** in the ear horn **308**. The mechanism uses the spring to maintain pressure on the pin **402** to maintain connection with the connector **408** when the connector **408** is in a position to lock the pin **402** in position. FIG. **5E** and FIG. **5F** illustrates a state where, after the ear horn **308** has been pulled into the state described in con-

nection with FIG. 5C and FIG. 5D, the ear horn 308 is rotated about the pin 402. This puts the ear horn 308 in a rotated position as described herein such that the first arm, with this rotated ear horn 308, does not interfere with the closure of the other arm 310 when the two arms are folded into the closed position. FIG. 5E and FIG. 5F illustrates the connector and pin assembly as continuing to secure the ear horn 308 to the temple 304 in the rotated position.

[0036] An aspect of the present invention relates to an adjustable nose bridge. An adjustable nose bridge may be important with head worn computers, especially those with computer displays, to ensure comfort and alignment of the displays and/or other portions of the head worn computer. FIG. 6 illustrates a HWC 102 with an adjustable nose bridge 602. The nose bridge is adjustable through a mechanism in the HWC 102. In embodiments, the mechanism includes a fixed notched attachment 604, a movable pin 608 adapted to fit into the notches of the notched attachment 604, and a selection device 610 that is attached to the movable pin 608. The movable pin 608 and nose bridge 602 are connected such that as the movable pin 608 shifts in position the nose bridge 602 moves in position as well. The selection device 610 causes the movable pin 608 to engage and disengage with the fixed notched attachment 604 when pressed and allowed to retract. As illustrated in FIG. 6, the selection device 610 is not in a pressed position so the movable pin 608 is engaged with the notched attachment 604 such that the nose bridge is securely attached in a stable position. FIG. 7 illustrates a scenario where the selection device is pressed, or activated, such that the moveable pin 608 is no longer engaged with the fixed notched attachment 604. This allows the nose bridge 602 to move up and down with respect to the rest of the HWC 102. Once the movable pin 608 aligns with a notch of the notched attachment 604, the two parts may engage to re-secure the nose bridge in the HWC 102.

[0037] In embodiments, a side arm of the HWC 102 may include an audio jack (not shown) and the audio jack may be magnetically attachable to the side arm. For example, the temple section 304 or ear horn section 308 may have a magnetically attachable audio jack with audio signal wires associated with an audio system in the HWC 102. The magnetic attachment may include one or more magnets on one end (e.g. on the head phone end or the side arm end) and magnetically conductive material on the other end. In other embodiments, both ends of the attachment may have magnets, of opposite polarization, to create a stronger magnetic bond for the headphone. In embodiments, the audio signal wires or magnetic connection may include a sensor circuit to detect when the headphone is detached from the HWC 102. This may be useful in situations where the wearer is wearing the headphones during a period when there is not constant audio processing (e.g. listening for people to talk with periods of silence). In embodiments, the other side's headphone may play a tone, sound, signal, etc. in the event a headphone is detached. In embodiments, an indication of the detachment may be displayed in the computer display.

[0038] In embodiments, the HWC 102 may have a vibration system that vibrates to alert the wearer of certain sensed conditions. In embodiments, the vibration system (e.g. an actuator that moves quickly to cause vibration in the HWC 102) may be mounted in a side arm (e.g. the temple section 304, or ear horn 308), in the top mount 312, etc. In embodiments, the vibration system may be capable of caus-

ing different vibration modes that may be indicative of different conditions. For example, the vibration system may include a multi-mode vibration system, piezo-electric vibration system, variable motor, etc., that can be regulated through computer input and a processor in the HWC 102 may send control signals to the vibration system to generate an appropriate vibration mode. In embodiments, the HWC 102 may be associated with other devices (e.g. through Bluetooth, WiFi, etc.) and the vibratory control signals may be associated with sensors associated with the other device. For example, the HWC 102 may be connected to a car through Bluetooth such that sensor(s) in the car can cause activation of a vibration mode for the vibration system. The car, for example, may determine that a risk of accident is present (e.g. risk of the driver falling asleep, car going out of its lane, a car in front of the wearer is stopped or slowing, radar in the car indicates a risk, etc.) and the car's system may then send a command, via the Bluetooth connection, to the HWC 102 to cause a vibratory tone to be initiated in the HWC102.

[0039] Another aspect of the present invention relates to a removable and replaceable speaker assembly for a HWC 102. There are times when different speaker types are desired or when a speaker may malfunction. It is therefore desirable to have a speaker assembly that is removable and replaceable by the user. To facilitate the removal and reattachment of the speaker assembly a magnetic or magnetic attachment system may be included. For example, the speaker assembly and head-worn computer may include magnetic elements such that the speaker can be removed by exerting pressure and replaced by getting the two sections close to one another. In another example, the speaker or head-worn computer may have a button, slider, etc. that can be interacted with to remove the speaker. In embodiments, the speaker assembly may have a form factor of an ear bud, ear phone, head phone, head set, external ear speaker, etc. In embodiments, the speaker assembly may include a vibratory system to provide haptic feedback to the user. In embodiments, such a removable and replaceable speaker system may be provided to both of the user's ears.

[0040] FIG. 8 illustrates several embodiments where HWC 102's are associated with speaker systems. Earbud 802 may be removably attached to the HWC 102 with a magnetic system or mechanical system or combination thereof. Speaker 804 may also be removably attached to the HWC102 in a similar way. The speaker 804 may be positioned to emit sound towards the user's ear but not actually be positioned in the ear. This configuration may provide for greater environmental hearing for the user as the ear would not be blocked by an ear bud, head phone, etc. The speaker 804 may generate audio waves and/or ultrasonic waves that are converted into audio when they are emitted through the air. When ultrasonic transducers are used, more than one frequency transducer may be included. See http://en.m.wikipedia.org/wiki/Sound_from_ultrasound and <http://www.holosonics.com> for references on generating sound from ultrasound. The speaker and/or piezo vibratory system 808 is depicted as integrated into the temple. In embodiments, this module may be integrated such that it can be removed and replaced and it may also be adapted such that it does not hang below the temple piece. Each of the removable and replaceable speaker systems described herein may include a vibratory system (e.g. piezo electric circuit that is controlled by the HWC 102).

[0041] In embodiments, a head-worn computer may include a temple portion mechanically secured to a computer display and adapted to position the computer display in front of an eye of a user, and the temple portion including a speaker attachment, wherein the speaker attachment is adapted to securely position a speaker assembly and electrically associate the speaker assembly with electronics internal to the head-worn computer and facilitate the user's release and re-securing of the speaker assembly with the temple portion. The speaker attachment may include a magnetic element, moveable mechanical element, etc. or combination thereof to secure and unsecure the speaker system from the HWC 102. The speaker assembly may have a form factor adapted to be inserted into an outer ear of the user, cover at least a portion of an outer ear of the user, cover substantially all of an outer ear of the user, to position the speaker under the temple assembly and above an ear of the user, to position a speaker under the temple assembly and in front of an ear of the user, angle the speaker towards the ear, etc. The speaker system may further have a vibratory system to provide haptic feedback to the user. The haptic feedback may be coordinated with a game being presented in the computer display, an application running on the HWC 102, etc. In embodiments, a vibratory system is provided in both speaker systems to provide controllable haptic feedback in stereo and/or on both or either side of the user's head.

[0042] In embodiments, the connection between the speaker system and the HWC 102 may be positioned other than under the temple section. It may be positioned on a side, top, bottom, end of a section of the side arm, for example. It may be positioned on the front bridge, for example. In embodiments, the speaker system may be connected to a top or side portion and the speaker may be further positioned to face forward, away from the user's ear. This may be a useful configuration for providing sound to others. For example, such a configuration may be used when the user wants to provide translations to a person nearby. The user may speak in a language, have the language translated, and then spoken through the forward facing speakers.

[0043] The removable nature of the speaker systems may be desirable for breakaway situations so a snag does not tear the glasses from the user or pull hard on the user's ear. The removable nature may also be useful for modularity configurations where the user wants to interchange speaker types or attach other accessories. For example, the user may want ear buds at one point and an open ear speaker configuration at another point and the user may be able to make the swap with ease given this configuration. The port on the HWC 102 may also be adapted for other accessories that include lights or sensors for example. The accessory may have an ambient light sensor to assist with the control of the lighting and contrast systems used in the HWC 102 displays, for example. In embodiments, the speaker port may be used as a charging port for the HWC 102 or data port for the HWC 102.

[0044] An aspect of the present invention relates to a language translation system with a head-worn computer. In embodiments, the head-worn computer includes a microphone and an audio communication system. The microphone is associated with a processor and is used to record a spoken utterance of a person wearing the head-worn computer. The microphone is positioned in the head-worn computer such that the person wearing the head-worn computer can speak in a normal tone and volume and the microphone

will record the user's utterances. The head-worn system is further adapted to convert the spoken utterance into a language other than a first spoken language represented by the spoken utterance, forming translated language. For example, if the user is speaking in English, the recorded English spoken words will be recognized and converted into a second language (e.g. Spanish). The converted language can then be played in an audio system such that the converted language is played through an associated speaker system.

[0045] The speaker in the language translation system may be wired or wirelessly connected to the head-worn computer. The speaker also will be of sufficient size and power to produce sound intensity such that a person at a conversational distance can hear the sound at a conversational level. For example, the speaker may produce sound in a range of approximately 30 to 40 db at a distance of 5 feet from the speaker. This is a relatively quiet conversational sound intensity and may be used in a quiet sound level environment. The speaker may produce sound in a range of approximately 40 to 60 db at a distance of 5 feet from the speaker. This is a moderate sound level and may be used in a moderate sound level environment. The speaker may produce sound in a range of approximately 60 to 80 db at a distance from 5 feet of the speaker. This is a relatively loud sound level and may be used in a loud environment. The speaker may be powerful enough for a maximum level (e.g. 80 db) and the audio system may be controllable to a lower level (e.g. 30 db). In embodiments, the sound level from the speaker may be automatically regulated depending on the sound in the environment. In embodiments, the sound level may be manually regulated. A manually regulated system may be regulated through gesture control, an external user interface adapted to regulate aspects of the head-worn computer, a control mounted on the head-worn computer, etc. The manually regulated system may be regulated by a control on the speaker itself.

[0046] In embodiments, the speaker is adapted to be positioned separately from the head-worn computer at a spot where it can be secured mechanically but to cause the sound to emanate from a direction similar to the direction of the user's speech. For example, the speaker may be mechanically adapted to fit into a shirt pocket of the user. This at least partially conceals the speaker and secures it such that the sound comes from a similar direction as the user's speech. The speaker may be mechanically adapted to clip onto a shirt of the user or otherwise adapted to be properly positioned.

[0047] The head-worn computer may be further adapted to communicate the translated language to the speaker and display, in a see-through head-worn display mechanically supported by the head-worn computer, a visual representation of the first or second language. For example, the visual representation may be a representation of the original language converted based on the utterances such that the person wearing the head-worn computer can see, as a confirmation, the language that was the output of the conversion process. This can help to avoid miscommunications. In the event that the conversion process improperly converted the spoken language, the user would be able to correct the system because the visual representations would give him the indication of improper conversion. As another example, the visual display may represent the converted language. This may help the person wearing the head-worn computer learn the second language. It may also be useful in a situation

where the user is trying to speak the second language and just wants help from the head-worn system from time to time. For example, the user may have the language conversion system off, and then turn it on when he finds himself lacking an understanding of how to say something in the second language. Once on, the conversion system can assist him by speaking the converted language to the other person and/or displaying the words or other representations in the see-through display.

[0048] In embodiments, the user may be able to turn the visual representation of the first or second language off. This user control may be helpful in a number of situations. For example, the user may want to rely on the conversion system and not want the additional distraction of the visual representations of the language presented in his field of view. This may be the case when the user needs to provide greater attention to the other person, situation or environment. The user control may be controlled by a gesture, external user interface, head-worn computer mounted user interface, etc. The user interface may also provide a visual representation of the user interface and/or the state of the user interface in the see-through display.

[0049] Another aspect of the language conversion system in the head-worn computer relates to the control of the audible tone of the second language produced through the speaker system. There may be situations where the use of a custom or predetermined audible tone may be useful. For example, if the audible tone from the speaker approximates the audible tone of the user himself than the system may produce a more personal situation between the user and the listener. For example, the translated language may have an audible spectrum based on the user's spoken utterances. The audible spectrum may be determined through an analysis of the spoken utterance's spectrum. The analysis may result in one of a plurality of pre-determined tones to be used when communicating the translated language to the speaker. For example, selected tone may be a high male tone, mid-tone male, low male tone, high female tone, mid-tone female, low female tone, etc. In embodiments, the user may manually select the tone.

[0050] In embodiments, the audible tone of the spoken utterances is analyzed and categorized by a signature, wherein the signature defines power within spectral ranges. The signature can then be used to modify the tone of the speaker system. For example, the signature may indicate that a high frequency range in the person's voice is not powerful so a spectral filter with a suppression of the range may be applied to the tone produced by the speaker system. A technique generally referred to as voice morphing may be used to approximate the user's voice. See the following published paper for additional information on voice morphing: Eng. Dept., Cambridge Univ., UK, DOI: 10.1109/ICASSP.2004.1325909 Conference: Acoustics, Speech, and Signal Processing, 2004. Proceedings. (ICASSP '04). IEEE International Conference on, Volume: 1.

[0051] Another aspect of the present invention relates to securing the head-worn computer **102** to the user's head in a way that the computer does not slip down the nose of the user, due to the extra front weight of the HWC **102**, but does not create discomfort for the user. While some have designed systems that use lateral force between the two side arms to squeeze the HWC arms on the sides of the user's head, this solution tends to cause comfort problems. The squeeze on the user's head has to be relatively high, as

compared to non-computer glasses, to maintain enough pressure to overcome the additional weight in the front of the glasses and this high pressure tends to cause comfort issues. In embodiments of the present invention, a substantially stiff ear horn is provided and the back end of the ear horn wraps around the user's head and touches the user's head. The touch point is towards the back of the user's head such that it provides a point or area of counteracting force for the HWC **102** if it tries to pull forward or down the user's nose due to the front weight of the HWC **102**. In embodiments, the end of the ear horn, or a section near the end, has a touch pad. The touch pad may be made of soft material so it is comfortable on the back of the user's head. In embodiments, the touch pad may be mounted such that it has angular flexibility. The angular flexibility allows the touch pad to better align with the touch point on the user's head so it can provide the counteractive force but spread the force over an area for greater comfort.

[0052] In embodiments, a head-worn computer is provided and has a see-through computer display configured to be mounted on the head of a user; a side arm configured to secure the see-through computer display to the user's head, the side arm further configured to be positioned to lay against the user's head proximate an ear of the user; and the side arm including a stiff member extending behind the ear of the user, contoured to substantially follow a curvature of the user's head behind the ear of the user, and to touch a portion of the user's head towards the rear of the user's head such that the see-through computer display remains substantially secure in a position in front of an eye of the user.

[0053] In embodiments, the stiff member is removeably secured to a temple portion of the side arm (as described herein elsewhere). The stiff member may be telescopically adjustable to fit the user's head. The stiff member may be provided with a ratchet style securing mechanism for adjusting the telescopic adjustment. The stiff member may be provided with a rotatable style securing mechanism for adjusting the telescopic adjustment, or another style securing mechanism may be provided. The stiff member may touch a portion of the user's head at a rear end of the stiff member. The rear end of the stiff member may include a touch pad. The touch pad may be made of a soft material to increase the comfort and surface area of the touch area. The touch pad may be attached such that it has angular flexibility such that the touch pad changes position to increase a touch surface in contact with the rear of the user's head.

[0054] FIG. 9 illustrates a HWC **102** mounted on the head of a user. The HWC **102** has a see-through optics module **204**, a temple portion **304**, a stiff ear horn **904** and a head-touch pad **902**. As described herein elsewhere, the stiff ear horn **904** may be removable and replaceable. This can be useful when the exchange of ear horns from one type to another or one size to another is desired, for example. The stiff ear horn **904** may be made of aluminum, aluminum tubing, carbon fiber, or other material that is relatively stiff. The stiffness should be of a level that provides for lateral inflexibility such that the touch pad **902** can exert counteracting force with a high rear facing vector. Too much flexibility in the stiff ear horn **904** can detract from the rear-facing vector of force when the weight of the HWC **102** is pulling forward/down the nose. In embodiments, several different lengths, shapes, stiffnesses, etc. of stiff ear horn **904** may be provided so the user can select the set that best serves his purpose. The head-touch pad **902** may be made of a soft

material, malleable material, etc. to provide comfort to the user and to increase the head touch surface. The head-touch pad **902** may also be mounted in such a way that the head-touch pad **902** can flex and/or change angle as it is pressed upon. The head-touch pad **902** may, for example, be mounted on the stiff ear horn **904** with a hinge or pivot mechanism such that the head-touch pad **902** self aligns with the user's head when the HWC **102** is put on the user's head. This configuration may increase the touch surface area between the head-touch pad **902** and the user's head and generate a larger counteracting force to prevent the slipping or moving of the HWC **102**.

[0055] In embodiments, the side arms of the HWC **102** are designed to exert inward lateral pressure on the user's head, but the lateral pressure is reduced so it is not uncomfortable, along with having stiff side arms **904** and head-touch pads **902**. In these embodiments, the ear horns **904** and head touch pads **902** cause significant counteracting forces in addition to the counteracting forces applied through the inward lateral forces applied by the side arms.

[0056] Another aspect of the present invention relates to the thermal management of the head-worn computer. In embodiments, the head-worn computer is a high functioning self-contained computer with wireless connectivity where the electronics are compacted to fit into a glasses style form factor. In embodiments, the main heat producing electronics are in an upper portion of a bridge section over the top of the lenses of the head-worn computer (e.g. as described in connection with FIG. 3B). The thermal management technologies described herein manage the heat such that the system can operate in high temperature environments (e.g. 55 degrees C. ambient) and maintain a comfortable temperature for the user (e.g. below 70 degrees C. along the forehead section).

[0057] FIG. 3B illustrates an embodiment of a head-worn computer wherein the electronics are housed in an upper bridge section above the lenses of the glasses form factor. This has been referred to as the top mount **312**. The top mount **312** may include the majority of the electronics that are used to form the fully functional computer. This may include the processor, memory, sensors, cameras, optical drive system, etc. In this embodiment, the batteries are housed in the temple portions of the side arms. The top mount **312** may include a cavity where the circuit board(s) are housed and secured. The top mount may also include a top plate **1000** designed to be mounted on the top of the cavity such that it forms a portion of the enclosure. In embodiments, the top plate **1000** is designed to receive heat from the circuit board and/or components mounted on the circuit board and then dissipate the heat into the environment surrounding the head-worn computer.

[0058] FIG. 10 illustrates several perspective views of an embodiment top plate **1000**. The top plate outside view **1002** illustrates the heat dissipating fins on the top of the top plate **1002**. The heat dissipating fins are on the outside of the completed head-worn computer assembly and dissipate the heat into the surrounding environment. The top plate front view **1004** illustrates a front perspective of the top plate. The top plate inside view **1008** illustrates a bottom view of the top plate. The bottom is the portion that is on the inside of the circuit board cavity in the fully assembled head-worn computer. The bottom of the top plate **1000** may have features to facilitate the mechanical and thermal connection with the circuit board and/or components on the circuit

board. For example, in the embodiment illustrated in FIG. 10 the top plate **1000** includes an internal component thermal connection pad **1010**. The thermal connection pad **1010** may be adapted to mechanically and thermally connect with the internal circuit board and/or a component on the circuit board (e.g. the processor, memory, or other heat producing component). The assembly may have intervening material between the top plate **1000** and the circuit board and/or circuit board component (e.g. a heat spreader plate designed to receive heat from a component or set of components and then spread the heat over an area, thermally conductive paste, glue, pad, or other facilitating material). In embodiments, a thermally conductive material is placed between the circuit board and/or circuit board component(s) and the thermal connection pad **1010** of the top plate **1000** to eliminate any air gap that might otherwise develop as a result of the mechanical mismatch of the components in the assembly. This can further facilitate the transfer of heat from the heat producing component(s) (e.g. the processor) to the thermal connection pad **1010** and out to the surrounding environment through the heat dissipating fins. The inventors used a thermally conductive material from Fujupoly in the thermally tested devices. This material has an advertised thermal conductivity of 11 Watt/m-k. Watt/m-k is a unit of measurement of thermal efficiency for thermal interface material.

[0059] FIG. 11 illustrates some test results relating to a head-worn computer with a top plate **1000** assembly as described herein. The head-worn computer was placed in an environmental chamber at 55 degrees Celsius. The head-worn computer was then turned on and the top plate **1000** temperature was observed. Operation of the processor was also observed. As can be seen, the maximum temperature of the top plate **1000** remained below 70 degrees Celsius. While 70 degrees Celsius is still a fairly hot maximum temperature, the heat dissipating fins caused the assembly to be comfortable to the human touch. In addition to maintaining an acceptable outside assembly temperature, the processor continued to operate throughout the testing, which is a significant advantage over the state of the art.

[0060] Although embodiments of HWC have been described in language specific to features, systems, computer processes and/or methods, the appended claims are not necessarily limited to the specific features, systems, computer processes and/or methods described. Rather, the specific features, systems, computer processes and/or methods are disclosed as non-limited example implementations of HWC.

[0061] All documents referenced herein are hereby incorporated by reference.

1. A system, comprising:
 - one or more processors;
 - a wearable device including a display, the wearable device configured to communicate with the one or more processors; and
 - a speaker configured to communicate with the one or more processors;
 wherein the one or more processors are configured to:
 - receive a spoken utterance of a user of the wearable device,
 - convert the spoken utterance into a translated utterance comprising a second language different from a first language of the spoken utterance,

- communicate the translated utterance to the speaker, and
display a visual representation of the translated utterance comprising the second language via the display.
2. The system of claim 1, wherein the one or more processors are further configured to communicate with the speaker via a wired connection.
3. The system of claim 1, wherein the one or more processors are further configured to wirelessly communicate with the speaker.
4. The system of claim 1, wherein the speaker is further configured to produce sound of less than 80 db at a distance of 5 feet from the speaker.
5. The system of claim 1, wherein the translated utterance comprises a spectral composition substantially similar to the spoken utterance.
6. The system of claim 5, wherein the one or more processors are further configured to:
determine the spectral composition via a spectral analysis of the spoken utterance, and
communicate the translated utterance in one of a plurality of voices to the speaker, the one of the plurality of voices determined based on the spectral analysis.
7. The system of claim 6, wherein the one voice comprises at least one of a high male voice, a mid-tone male voice, a low male voice, a high female voice, a mid-tone female voice, and a low female voice.
8. The system of claim 1, wherein the speaker is mechanically adapted to fit into a pocket of clothing.
9. The system of claim 1, wherein the speaker is mechanically adapted to clip onto clothing.
10. The system of claim 1, wherein the one or more processors are further configured to display the visual representation via a graphical user interface.
11. The system of claim 1, wherein the one or more processors are further configured to display a visual representation of the spoken utterance comprising the first language via the display.
12. The system of claim 11, wherein the one or more processors are further configured to switch from the display

of the visual representation of the translated utterance comprising the second language to the display of the visual representation of the spoken utterance comprising the first language.

13. The system of claim 1, wherein the one or more processors are further configured to:
receive an input from the user of the wearable device, and
in response to receiving the input, forgo displaying the visual representation of the translated utterance.
14. The system of claim 1, wherein the one or more processors are further configured to, in response to receiving a signal, forgo displaying the visual representation of the translated utterance.
15. The system of claim 1, wherein the wearable device comprises a wearable head device.
16. A method comprising:
receiving a spoken utterance of a user of a wearable device,
converting the spoken utterance into a translated utterance comprising a second language different from a first language of the spoken utterance,
communicating the translated utterance to a speaker, and
displaying a visual representation of the translated utterance comprising the second language via a display of the wearable device.
17. The method of claim 16, further comprising presenting a graphical user interface that displays the visual representation of the translated utterance comprising the second language.
18. The method of claim 16, further comprising switching from the display of the visual representation of the translated utterance comprising the second language to display of a visual representation of the spoken utterance comprising the first language.
19. The method of claim 16, further comprising in response to receiving a signal, forgoing displaying the visual representation of the translated utterance.
20. The method of claim 16, wherein the wearable device comprises a wearable head device.

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